

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I DOMA SAKETH-192525018____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- A. Explain why this issue occurs.
- B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

1. Secure Data Transmission using Block Cipher Scenario

A company is transmitting financial data such as customer account numbers, transaction amounts, account balances, and payment details over a network. Many of these records contain repeated values. Initially, the company uses **Electronic Codebook (ECB)** mode for encryption. After monitoring the encrypted data, the company notices that similar patterns appear in the ciphertext, raising concerns about data confidentiality.

A. Explain why this issue occurs.

Electronic Codebook (ECB) Mode

ECB is the simplest block cipher mode. In ECB mode, the plaintext is divided into fixed-size blocks, and each block is encrypted independently using the same encryption key.

The encryption process is:

where:

- = Plaintext block
- = Ciphertext block
- = Encryption using key

Since every block is encrypted independently, identical plaintext blocks always produce identical ciphertext blocks.

Example

Suppose the company sends the following blocks:

Plaintext Block Ciphertext Block (ECB)

Transaction A X1A3

Transaction A X1A3

Transaction B B7F9

Transaction A X1A3

Notice that the repeated plaintext **Transaction A** always produces the same ciphertext **X1A3**.

An attacker cannot directly read the encrypted data but can easily recognize repeated ciphertext patterns.

Why is this a security problem?

When financial records contain repeated information such as:

- Account numbers
- Customer IDs
- Fixed transaction headers
- Salary payments
- Bank identifiers

ECB produces repeated ciphertext blocks.

An attacker observing network traffic may discover:

- Frequently occurring transactions
- Number of repeated records
- Data structure
- Relationships between encrypted messages

Although the attacker cannot immediately decrypt the information, these patterns may help in statistical analysis or cryptanalysis.

ECB therefore leaks structural information about the plaintext.

Illustration

Suppose four plaintext blocks are:

P1 = Salary

P2 = Salary

P3 = Bonus

P4 = Salary

ECB encryption becomes

Salary → A3B1

Salary → A3B1

Bonus → D7E4

Salary → A3B1

Ciphertext:

A3B1

A3B1

D7E4

A3B1

The repeated ciphertext clearly reveals that the same data occurred multiple times.

Disadvantages of ECB

- Reveals repeated data patterns.
- Does not hide similarities in plaintext.
- Vulnerable to traffic analysis.
- Not suitable for confidential financial information.
- Does not provide semantic security.

Conclusion

ECB mode encrypts each plaintext block independently using the same key.

Therefore, identical plaintext blocks always generate identical ciphertext blocks, exposing patterns that attackers can observe. For sensitive financial transactions, ECB is considered insecure because it fails to hide the structure of the original data.

B. Suggest a more secure block cipher mode and justify your choice.

Recommended Mode: Cipher Block Chaining (CBC)

Cipher Block Chaining (CBC) is a much more secure block cipher mode than ECB. Instead of encrypting each block independently, CBC combines each plaintext block with the previous ciphertext block before encryption.

The first block uses an **Initialization Vector (IV)**.

Encryption formula:

where

-
- = Plaintext block
- = Ciphertext block

Working of CBC

Suppose

IV = 0000

Plaintext

Salary

Salary

Bonus

Salary

Even though "Salary" appears multiple times, each occurrence is first XORed with the previous ciphertext.

As a result:

Salary → X1D4

Salary → A9F2

Bonus → H7P8

Salary → Q2L5

Notice that every ciphertext block is different.

An attacker can no longer determine whether the plaintext blocks are identical.

Advantages of CBC

1. Hides Data Patterns

Repeated plaintext blocks generate different ciphertext blocks.

2. Improved Confidentiality

Each ciphertext depends on both the plaintext and the previous ciphertext block.

3. Resistant to Pattern Analysis

Attackers cannot identify repeated records.

4. Suitable for Financial Systems

CBC protects confidential information such as:

- Banking transactions
- Salary records
- Payment details
- Customer account information

Comparison of ECB and CBC

Feature	ECB	CBC
Encrypts blocks independently	Yes	No
Uses Initialization Vector	No	Yes
Reveals plaintext patterns	Yes	No
Suitable for repeated financial data	No	Yes
Security Level	Low	High

Conclusion

CBC is the preferred block cipher mode because it eliminates the pattern leakage present in ECB mode. By using an Initialization Vector and chaining each block with the previous ciphertext, CBC ensures that identical plaintext blocks produce different

ciphertext blocks, making it highly suitable for transmitting confidential financial information securely.

2. Choosing an Encryption Algorithm for Banking System

Scenario

A banking application processes sensitive customer information, including:

- Account numbers
- Debit and credit card details
- Online fund transfers
- Payment information
- Customer passwords
- Financial records

The system requires:

- High security
- Moderate encryption speed
- Reliable protection against attacks

Available encryption algorithms:

- DES
- 3-DES
- Blowfish

A. Identify the most suitable algorithm.

Answer

3-DES (Triple Data Encryption Standard) is the most suitable encryption algorithm for this banking application.

B. Justify your choice based on security strength and efficiency.

1. DES (Data Encryption Standard)

Key Size

56 bits

Working

DES encrypts data using a 56-bit secret key.

Although DES was once considered secure, advances in computing have made brute-force attacks practical.

Advantages

- Simple implementation
- Fast encryption

Disadvantages

- Small key size
- Easily broken by modern computers
- Not suitable for banking applications

Conclusion: DES should not be used because it does not provide sufficient security for sensitive customer transactions.

2. 3-DES (Triple DES)

Key Size

168 bits

Working

3-DES performs the DES encryption process three times:

- Encrypt
- Decrypt
- Encrypt

This significantly increases the difficulty of brute-force attacks.

Advantages

- Much stronger security than DES

- Well-tested and reliable
- Historically used in banking and ATM systems
- Resistant to brute-force attacks

Disadvantages

- Slower than DES
- Higher computational cost

Despite being slower, the increased security makes it suitable for protecting financial transactions.

3. Blowfish

Key Size

Up to 448 bits (commonly 128 bits or higher)

Working

Blowfish is a symmetric block cipher designed for high speed and strong security.

Advantages

- Very fast encryption
- Strong security
- Flexible key size
- Low memory requirements

Disadvantages

- Less commonly used in modern banking standards
- Replaced in many systems by AES

Comparison

Feature	DES	3-DES	Blowfish
Key Size	56 bits	168 bits	128–448 bits
Security	Low	Very High	High
Encryption Speed	Fast	Moderate	Very Fast
Resistance to Brute Force	Poor	Excellent	Excellent
Suitable for Banking	No	Yes	Yes
Overall Performance	Good	Moderate	Excellent

Why 3-DES is the Best Choice

A banking application prioritizes **security over speed**.

Customer financial information must remain confidential even if encryption takes slightly longer.

3-DES offers:

- Strong encryption through triple encryption operations.
- High resistance to brute-force attacks.
- Proven reliability in banking, ATM networks, and financial institutions.
- Moderate performance that is acceptable for secure customer transactions.

Although Blowfish is faster, the scenario specifically emphasizes **high security with moderate performance**, making 3-DES the most appropriate choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand